

A negative answer to a question of Carenini on almost independent sets in regular graphs

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Abstract

For a d -regular graph G on n vertices and $\gamma \geq 0$, let $i_\gamma(G)$ be the number of vertex sets that span at most γdn edges. Carenini asked whether, when $2d \mid n$, the disjoint union of $n/(2d)$ copies of $K_{d,d}$ maximises i_γ among d -regular graphs on n vertices for every $\gamma \geq 0$. For $\gamma = 0$ this is the theorem of Kahn and Zhao on independent sets. We show that the answer is no. The smallest counterexample (smallest n , and smallest γ for that n) is $(n, d, \gamma) = (6, 3, 1/18)$: the triangular prism has 28 vertex sets spanning at most one edge, and $K_{3,3}$ has 24. More generally, for every pair (n, d) with $n > 0$ and $2d \mid n$, except the trivial cases $d = 1$ and $(n, d) = (4, 2)$ in which only one d -regular graph exists, the answer is negative for some $\gamma < 1/2$; for $d \geq 3$ the union of copies of $K_{d,d}$ is then even a minimiser, and every d -regular graph containing a triangle beats it. For small γ we give two families: a switched $K_{d,d}$ beats $K_{d,d}$ on $2d$ vertices for every $d \geq 3$, and a connected bipartite d -regular graph beats two copies of $K_{d,d}$ on $4d$ vertices for every $d \geq 2$, so the answer is no even among bipartite graphs. For every fixed $\gamma \in (1/8, 1/2)$ and every $d \geq 2$, a bipartite competitor wins for all sufficiently large n . On the other hand, the union of copies of $K_{d,d}$ is optimal at the exponential scale for every γ ; the upper bound here is Carenini's biclique reduction, which rests on a theorem of Sah, Sawhney, Stoner and Zhao. The exact question remains open for fixed $\gamma \in (0, 1/8]$ as $n \rightarrow \infty$, and our counterexamples are consistent with the weaker statement in which the union of copies of $K_{d,d}$ is optimal up to a factor $1 + o(1)$. The counterexample at $(6, 3, 1/18)$, the switched family for every $d \geq 3$ and three further instances are formally verified in Lean 4. The counterexamples and proofs were found by AI coding agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

Background. For a graph G and a set $A \subseteq V(G)$ let $e_G(A)$ be the number of edges of G with both ends in A . Following Carenini [3], for a d -regular graph G on n vertices and a real number $\gamma \geq 0$ put

$$i_\gamma(G) = |\{A \subseteq V(G) : e_G(A) \leq \gamma dn\}|,$$

so that $i_0(G)$ is the number of independent sets of G . Alon [1] proved that a d -regular graph on n vertices has at most $2^{(1/2+o(1))n}$ independent sets, where $o(1) \rightarrow 0$ as $d \rightarrow \infty$. Kahn [8] proved the sharp bound $i_0(G) \leq (2^{d+1} - 1)^{n/(2d)}$ for bipartite d -regular graphs, and Zhao [16] extended it to all d -regular graphs. Since $K_{d,d}$ has $2^{d+1} - 1$ independent sets, this says that when $2d \mid n$, the disjoint union $mK_{d,d}$ of $m = n/(2d)$ copies of $K_{d,d}$ has the largest number of independent sets among d -regular graphs on n vertices.

In work on tolerant testing of independent sets, Seth [14] bounded the number of sets $J \subseteq V(G)$ whose edge density $e_G(J)/\binom{|J|}{2}$ is at most $d/(kn)$, and asked for the correct order

t	0	1	2	3	4	5	6	7	8	9
$N_{\leq t}(K_{3,3})$	15	24	42	48	57	57	63	63	63	64
$N_{\leq t}(K_3 \square K_2)$	13	28	40	48	57	57	63	63	63	64

Table 1: The number of vertex sets spanning at most t edges, for the two cubic graphs on six vertices. Here $i_\gamma = N_{\leq t}$ with $t = \lfloor 18\gamma \rfloor$.

of this bound and whether the disjoint union of $n/(2d)$ copies of $K_{d,d}$ is extremal. Carenini [3] proved a robust form of the Kahn–Zhao theorem for the count i_γ : there is an absolute constant C such that

$$i_\gamma(G) \leq 2^{n/2} \exp\{C\gamma \log(e/\gamma) n + Cn/d\}$$

for every d -regular graph G on n vertices with $d \geq 2$ and every $0 < \gamma \leq 1/2$ [3, Theorem 1.1], and both error terms are needed [3, Proposition 1.2]. As Carenini notes, Seth’s density condition implies $e_G(J) \leq dn/(2k)$, so this answers the quantitative part of Seth’s question. The key step [3, Proposition 2.1] is a biclique reduction based on a theorem of Sah, Sawhney, Stoner and Zhao [13, Corollary 1.15] (Corollary 1.14 in version 2 of the arXiv preprint, the numbering used in [3]): for $0 < q \leq 1$, the sum of $q^{e_G(A)}$ over all $A \subseteq V(G)$ is at most the $n/(2d)$ -th power of the sum of $q^{e(A)}$ over all subsets A of $V(K_{d,d})$. Carenini then asked for the exact extremal statement, which we restate in our notation.

Question ([3, Question 1.3]). *Suppose that $2d \mid n$, and let $m = n/(2d)$. Is it true that for every $\gamma \geq 0$, every d -regular graph G on n vertices satisfies $i_\gamma(G) \leq i_\gamma(mK_{d,d})$?*

For $\gamma = 0$ this is the theorem of Kahn and Zhao. For $\gamma \geq 1/2$ it holds trivially, since every set spans at most $dn/2 \leq \gamma dn$ edges and so $i_\gamma(G) = 2^n$. We call a pair (n, d) *admissible* if $d \geq 1$, $n \geq 1$ and $2d \mid n$, that is, if n is a positive multiple of $2d$. Seth’s count differs from i_γ , because its density condition depends on $|J|$; our results concern Question 1.3, and we make no claim about Seth’s count.

Results. For an integer t let $N_{\leq t}(G)$ be the number of sets $A \subseteq V(G)$ with $e_G(A) \leq t$. Since $e_G(A)$ is an integer, $i_\gamma(G) = N_{\leq \lfloor \gamma dn \rfloor}(G)$. Hence an inequality $N_{\leq t}(G) > N_{\leq t}(mK_{d,d})$ at an integer $t \geq 0$ gives a negative answer to Question 1.3 for every $\gamma \in [t/(dn), (t+1)/(dn))$. We write $T(G)$ for the number of triangles of G , C_k for the cycle of length k , and $G \square H$ for the Cartesian product.

Theorem A. *Let $d \geq 2$, let n be a positive multiple of $2d$ with $(n, d) \neq (4, 2)$, let $m = n/(2d)$, and put*

$$t^* = \frac{dn}{2} - 3d + 1, \quad \gamma^* = \frac{t^*}{dn} = \frac{1}{2} - \frac{3d-1}{dn}.$$

Then $t^ \geq 1$, and the following hold.*

- (a) *If $d \geq 3$, every d -regular graph G on n vertices satisfies $i_{\gamma^*}(G) - i_{\gamma^*}(mK_{d,d}) \geq \frac{7}{5}T(G)$, and $i_{\gamma^*}(G) = i_{\gamma^*}(mK_{d,d})$ if and only if G is triangle-free. So $mK_{d,d}$ minimises i_{γ^*} among d -regular graphs on n vertices, and every such graph that contains a triangle has a larger value.*
- (b) *If $d = 2$, every 2-regular graph $G \neq mC_4$ on n vertices satisfies $i_{\gamma^*}(G) > i_{\gamma^*}(mC_4)$.*

The excluded cases are trivial: for $d = 1$ the only 1-regular graph on n vertices is $\frac{n}{2}K_2 = \frac{n}{2}K_{1,1}$, and $C_4 = K_{2,2}$ is the only 2-regular graph on four vertices. For every other admissible pair there is a competitor. If $d \geq 3$, the graph $(K_d \square K_2) \cup (m-1)K_{d,d}$ is d -regular and contains a triangle; if $d = 2$ and $n \geq 8$, the cycle C_n differs from mC_4 . So the answer to Question 1.3 is negative for every admissible pair (n, d) except the trivial ones.

For $(n, d) = (6, 3)$ we get $t^* = 1$ and $\gamma^* = 1/18$. The only cubic graphs on six vertices are $K_{3,3}$ and the triangular prism $K_3 \square K_2$, and Table 1 lists their counts. The prism has 28 sets spanning at most one edge, and $K_{3,3}$ has 24. So $K_{3,3}$ fails to be a maximiser exactly for $\gamma \in [1/18, 1/9)$. Since $n = 6$ is the smallest order at which Question 1.3 is not trivial, $(n, d, \gamma) = (6, 3, 1/18)$ is the smallest counterexample, in the sense that it has the smallest n and, for that n , the smallest γ .

For small γ we give two explicit families.

Theorem B. *Let $d \geq 3$, let $K_{d,d}$ have sides $X = \{x_1, \dots, x_d\}$ and $Y = \{y_1, \dots, y_d\}$, and let S_d be obtained from $K_{d,d}$ by deleting the edges x_1y_1 and x_2y_2 and adding the edges x_1x_2 and y_1y_2 . Then S_d is d -regular, every set spanning at most one edge of $K_{d,d}$ spans at most one edge of S_d , and*

$$N_{\leq 1}(S_d) = N_{\leq 1}(K_{d,d}) + 4d - 8 = 2^{d+1} + d^2 + 4d - 9.$$

Consequently, for $n = 2d$ and every $\gamma \in [1/(2d^2), 1/d^2)$, $K_{d,d}$ does not maximise i_γ among d -regular graphs on n vertices.

The graph S_3 is the prism. Corollary 5 gives an explicit criterion for $S_d \cup (m-1)K_{d,d}$ to beat $mK_{d,d}$.

Theorem B'. *Let $d \geq 2$. Take two disjoint copies (X_1, Y_1) and (X_2, Y_2) of $K_{d,d}$, choose $u_k \in X_k$ and $w_k \in Y_k$ for $k = 1, 2$, delete the edges u_1w_1 and u_2w_2 , and add the edges u_1w_2 and u_2w_1 . The resulting graph H_d is d -regular, bipartite and connected, and*

$$N_{\leq 1}(H_d) - N_{\leq 1}(2K_{d,d}) = 2(d-1)(2^d + d - 3) > 0, \quad i_0(H_d) - i_0(2K_{d,d}) = -2(2^{d-1} - 1)^2.$$

Consequently, for $n = 4d$ and every $\gamma \in [1/(4d^2), 1/(2d^2))$, $2K_{d,d}$ does not maximise i_γ , even among bipartite d -regular graphs on n vertices.

So the answer is negative even within the bipartite class for which Kahn proved the case $\gamma = 0$. The graph H_2 is the cycle C_8 .

At the exponential scale, on the other hand, the answer is positive for every γ . For $q > 0$ put

$$P_d(q) = \sum_{a,b=0}^d \binom{d}{a} \binom{d}{b} q^{ab},$$

which is the sum of $q^{e(A)}$ over all subsets A of $V(K_{d,d})$.

Proposition C. *Let $d \geq 2$ and $\gamma \geq 0$, and put*

$$\rho(d, \gamma) = \inf_{0 < q \leq 1} \left(\frac{1}{2d} \log P_d(q) - \gamma d \log q \right).$$

Every d -regular graph G on n vertices satisfies $\frac{1}{n} \log i_\gamma(G) \leq \rho(d, \gamma)$, and $\frac{1}{n} \log i_\gamma(mK_{d,d}) \rightarrow \rho(d, \gamma)$ as $n \rightarrow \infty$ through multiples of $2d$, where $m = n/(2d)$.

The upper bound is Carenini's Proposition 2.1 with the fugacity equal to 1, optimised over q . Proposition C shows that the maximum of i_γ over d -regular graphs on n vertices is $i_\gamma(mK_{d,d}) e^{o(n)}$ for fixed d and γ ; all the failures in this paper are subexponential. They are not confined to small n , however.

Theorem D. *Let $d \geq 2$ and $1/8 < \gamma < 1/2$. For n divisible by $2d$ write $n/(2d) = 2k + r$ with $r \in \{0, 1\}$, and let $G_n = kH_d \cup rK_{d,d}$, a bipartite d -regular graph on n vertices. Then $i_\gamma(G_n) > i_\gamma(mK_{d,d})$, where $m = n/(2d)$, for all sufficiently large n divisible by $2d$.*

The proof compares the upper tails of the two counts with Cramér's theorem in its qualitative form, so it gives no explicit threshold n_0 beyond which the inequality holds.

Scope. Our counterexamples are exact-count effects. In Theorems B and B' the threshold is a single edge, $\lfloor \gamma dn \rfloor = 1$. In Theorem A, γ^* exceeds $1/8$ unless $(n, d) = (6, 3)$, and $\gamma^* \rightarrow 1/2$ as $n \rightarrow \infty$. In Theorem D, $\gamma > 1/8$ is fixed; both counts are then $(1 - o(1))2^n$, and their relative difference tends to 0 (Remark 10). The value $1/8$ is where the mean crosses the threshold: a uniformly random subset of a d -regular graph on n vertices spans $dn/8$ edges on average. Question 1.3 remains open for fixed $\gamma \in (0, 1/8]$ as $n \rightarrow \infty$. All our counterexamples with $\gamma \leq 1/8$ have $\gamma d < 1$. In particular, we know no counterexample with $\gamma \leq 1/8$ and $\gamma d \geq 2$, the range of small γ in which Carenini's Proposition 1.2 shows that the term $\gamma \log(e/\gamma)n$ of her Theorem 1.1 is needed. Our counterexamples also do not refute the weaker statement $i_\gamma(G) \leq (1 + o(1))i_\gamma(mK_{d,d})$. Section 2 lists several readings of the question with our status for each, and Section 10 states open questions.

Prior work. Question 1.3 appeared in version 1 of [3], submitted to the arXiv on 22 September 2026. We found no earlier answer to it, and no discussion of it, in the following sources, which we searched on 26 September 2026 between 05:12 and 05:23 UTC: the arXiv (the listing of [3], which had only version 1, and searches of the arXiv API for later papers on the topic or by the same author); Semantic Scholar, which listed no citations of [3]; the problem database MathDB (<https://mathdb.com>); GitHub searches of code, commits, issues, pull requests and repositories; the public repositories of AI-assisted work on open problems that we check, among them the `trureturing` repository [15]; and web search. Some of these searches were repeated on 26 September 2026 between 06:39 and 06:42 UTC, with the same result: we searched the issues and pull requests of the `trureturing` repository for the versioned identifier 2609.28527v1 (as a positive control, the same search for the versioned identifier of another recent arXiv paper found the expected issue), searched GitHub issues and commits, and searched MathDB by title. We did not search Google Scholar, zbMATH or MathOverflow. This reports the coverage of our searches, not a guarantee of priority. Theorem 1.1 and Proposition 2.1 of [3] are due to Carenini. The upper bound in Proposition C is a direct consequence of the latter, which rests on the theorem of Sah, Sawhney, Stoner and Zhao [13, Corollary 1.15].

Organisation. Section 2 discusses readings of the question. Section 3 collects notation and a complement identity. Theorem A is proved in Section 4, Theorems B and B' in Section 5, Proposition C in Section 6 and Theorem D in Section 7. Section 8 reports computations, Section 9 the formal verification, and Section 10 open questions.

2 Readings of the question

Question 1.3 is a single universal statement, but it can be weakened in several natural ways. Table 2 lists the readings we considered; R1 is the literal one. In R4–R6, d and γ are fixed and $n \rightarrow \infty$ through multiples of $2d$.

	Reading	Status
R1	Literal: for all admissible (n, d) and all $\gamma \geq 0$, $i_\gamma(G) \leq i_\gamma(mK_{d,d})$ for every d -regular graph G on n vertices.	False. Smallest counterexample (smallest n , then smallest γ): $(n, d, \gamma) = (6, 3, 1/18)$ (Theorems A and B); formally verified.
R2	For a given admissible pair (n, d) , the inequality holds for all $\gamma \geq 0$.	False for every admissible pair with $d \geq 2$ and $(n, d) \neq (4, 2)$ (Theorem A); true in the trivial cases $d = 1$ and $(n, d) = (4, 2)$.
R3	Small γ : the inequality holds for all admissible (n, d) and all $0 \leq \gamma < 1/8$.	False: for $n = 2d$ and every $d \geq 3$ (Theorem B); for $n = 4d$ and every $d \geq 2$, with a connected bipartite competitor (Theorem B'); for some $n = 2dm$ with $m \geq 2$ (Corollary 5); at $\gamma = 1/10$ for many $n \leq 400$ (Remark 12).
R4	Fixed γ : the inequality holds for all large n .	False for every $d \geq 2$ and every $\gamma \in (1/8, 1/2)$ (Theorem D); true for $\gamma = 0$ and for $\gamma \geq 1/2$.
R5	R4 for fixed $\gamma \in (0, 1/8]$.	Open. True at the exponential scale (Proposition C). The competitors of Theorem D lose for large n when $\gamma < 1/8$ (Remark 11).
R6	Ratio form: $\max_G i_\gamma(G) \leq (1 + o(1)) i_\gamma(mK_{d,d})$.	Not refuted. True for $\gamma = 0$; for $\gamma = 1/8$, where $i_{1/8}(G) = (1/2 + o(1))2^n$ uniformly in G ; and for $\gamma > 1/8$, where $i_\gamma(mK_{d,d}) = (1 - o(1))2^n$ (Remark 10). Open for $0 < \gamma < 1/8$.

Table 2: Readings of Question 1.3 of [3]; here $m = n/(2d)$.

3 Preliminaries

All graphs are finite and simple. For a graph G let $N_s(G)$ be the number of sets $A \subseteq V(G)$ with $e_G(A) = s$, and let

$$P_G(z) = \sum_{A \subseteq V(G)} z^{e_G(A)} = \sum_{s \geq 0} N_s(G) z^s,$$

so that $N_{\leq t}(G) = \sum_{s \leq t} [z^s] P_G(z)$ and $i_0(G) = N_0(G)$, where $[z^s] F(z)$ denotes the coefficient of z^s in a polynomial F . In the notation of [3], $P_G(q) = Z_G(1, q)$. For a disjoint union, $P_{G \cup F} = P_G P_F$, since $e_{G \cup F}(A) = e_G(A \cap V(G)) + e_F(A \cap V(F))$. Choosing a vertices on one side of $K_{d,d}$ and b on the other spans ab edges, so $P_{K_{d,d}} = P_d$ and $P_{mK_{d,d}} = P_d^m$. For a d -regular graph G on n vertices we write $E = e(G) = dn/2$. Every integer $t \geq 0$ arises as $\lfloor \gamma dn \rfloor$, namely for $\gamma \in [t/(dn), (t+1)/(dn))$, so Question 1.3 is equivalent to the inequalities $N_{\leq t}(G) \leq N_{\leq t}(mK_{d,d})$ for all integers $t \geq 0$.

Lemma 1 (complement identity). *Let G be a d -regular graph with n vertices and E edges, and for $S \subseteq V(G)$ let $\partial(S)$ be the number of edges meeting S . Then $\partial(S) = d|S| - e_G(S)$, and for every integer t ,*

$$N_{\leq t}(G) = 2^n - |\{S \subseteq V(G) : \partial(S) \leq E - t - 1\}|.$$

Proof. Summing the degrees of the vertices of S counts each edge inside S twice and each edge between S and its complement once, so $d|S| = e_G(S) + \partial(S)$. The edges inside $V(G) \setminus S$ are exactly the edges not meeting S , so $e_G(V(G) \setminus S) = E - \partial(S)$. Hence $e_G(A) \leq t$ if and only if $\partial(V(G) \setminus A) \geq E - t$, and $A \mapsto V(G) \setminus A$ is a bijection. $\square$

4 Proof of Theorem A

Proposition 2. *Let $d \geq 2$, let G be a d -regular graph on n vertices, and let $t^* = dn/2 - 3d + 1$. Then*

$$N_{\leq t^*}(G) = 2^n - 1 - n - \binom{n}{2} - n \binom{d}{2} + 2T(G) - Q(G),$$

where $Q(G)$ is the number of 4-sets spanning at least $d + 2$ edges plus the number of 5-sets spanning at least $2d + 2$ edges.

Proof. We have $E - t^* - 1 = 3d - 2$. By Lemma 1, $2^n - N_{\leq t^*}(G)$ is the number of sets S with $\partial(S) = d|S| - e_G(S) \leq 3d - 2$. We count these sets by their size k .

- $k \leq 2$: every set qualifies, since $\partial(S) \leq kd \leq 2d \leq 3d - 2$. This gives $1 + n + \binom{n}{2}$ sets.
- $k = 3$: the condition is $e_G(S) \geq 2$. Call a pair of edges with a common end a *cherry*; there are $n \binom{d}{2}$ cherries. A 3-set with exactly two edges contains exactly one cherry, and a triangle contains three. Hence there are $n \binom{d}{2} - 3T(G)$ sets of size 3 with exactly two edges, and $n \binom{d}{2} - 2T(G)$ with at least two.
- $k = 4$ and $k = 5$: the conditions are $e_G(S) \geq d + 2$ and $e_G(S) \geq 2d + 2$, so these sets are counted by $Q(G)$.
- $k \geq 6$: we would need $e_G(S) \geq (k - 3)d + 2$, while $2e_G(S) \leq kd$ as every vertex has degree d . Together these give $(k - 6)d \leq -4$, which is impossible.

Adding up proves the formula. $\square$

Lemma 3. *Let G be a d -regular graph. If $d \geq 5$ then $Q(G) = 0$; if $d = 4$ then $Q(G) \leq \frac{3}{5}T(G)$; if $d = 3$ then $Q(G) \leq \frac{1}{2}T(G)$; and if $d = 2$ then $Q(G)$ is the number $c_4(G)$ of components of G isomorphic to C_4 . In particular, if $d \geq 3$ then $2T(G) - Q(G) \geq \frac{7}{5}T(G)$.*

Proof. If $d \geq 5$, a 4-set spans at most $6 < d + 2$ edges and a 5-set spans at most $10 < 2d + 2$ edges.

Let $d = 4$. Then $Q(G)$ is the number of copies of K_4 plus the number of copies of K_5 . A triangle xyz lies in at most two copies of K_4 , because the fourth vertex is a neighbour of x other than y and z , and x has only two such neighbours. As each K_4 contains four triangles, the number of copies of K_4 is at most $2T(G)/4$. Each vertex of a K_4 has exactly one neighbour outside it, so a K_4 lies in at most one K_5 . As each K_5 contains five copies of K_4 , the number of copies of K_5 is at most a fifth of the number of copies of K_4 . Hence $Q(G) \leq \frac{6}{5} \cdot \frac{T(G)}{2} = \frac{3}{5}T(G)$.

Let $d = 3$. A 5-set spans at most $\lfloor 15/2 \rfloor = 7 < 8$ edges, so $Q(G)$ counts the 4-sets with at least five edges. Each of them contains at least two triangles. A triangle xyz lies in at most one of them: its fourth vertex must be adjacent to at least two of x, y, z ; if w and w' both have this property, then some vertex of $\{x, y, z\}$ is adjacent to both, and since that vertex has only one neighbour outside $\{x, y, z\}$, $w = w'$. Counting pairs consisting of a triangle and a set counted by $Q(G)$ that contains it gives $2Q(G) \leq T(G)$.

Let $d = 2$. A 4-set with four edges induces a 2-regular graph on four vertices, that is, a component C_4 ; and a 5-set spans at most five edges.

The last assertion follows because $\frac{3}{5} \geq \frac{1}{2}$. $\square$

Proof of Theorem A. First, $t^* \geq 1$: if $d \geq 3$, then $n \geq 2d$ gives $t^* \geq d^2 - 3d + 1 \geq 1$; if $d = 2$, then $n \geq 8$ and $t^* = n - 5 \geq 3$. As $\gamma^* dn = t^*$, we have $i_{\gamma^*} = N_{\leq t^*}$. By Proposition 2, for d -regular graphs G and G' on n vertices,

$$N_{\leq t^*}(G) - N_{\leq t^*}(G') = (2T(G) - Q(G)) - (2T(G') - Q(G')).$$

The graph $mK_{d,d}$ is bipartite, so $T(mK_{d,d}) = 0$. If $d \geq 3$, then also $Q(mK_{d,d}) = 0$ by Lemma 3, and (a) follows from Lemma 3: the difference $2T(G) - Q(G)$ is at least $\frac{7}{5}T(G)$, and it vanishes when $T(G) = 0$. If $d = 2$, Lemma 3 gives $N_{\leq t^*}(G) - N_{\leq t^*}(mC_4) = 2T(G) + n/4 - c_4(G)$, which is positive unless $T(G) = 0$ and $c_4(G) = n/4$, that is, unless $G = mC_4$. $\square$

Remark 4. For $n = 2d$, every d -regular graph $G \neq K_{d,d}$ contains a triangle. Indeed, if G is triangle-free and uv is an edge, the neighbourhoods $N(u)$ and $N(v)$ are disjoint independent sets of size d , so they partition $V(G)$; every vertex of $N(u)$ then has all its d neighbours in $N(v)$, and $G = K_{d,d}$. (This is also the equality case of Mantel's theorem [9], since G has d^2 edges.) Hence, for every $d \geq 3$, every d -regular graph on $2d$ vertices other than $K_{d,d}$ beats $K_{d,d}$ at $t^* = d^2 - 3d + 1$. As a second example, for $(n, d) = (12, 3)$ we have $t^* = 10$ and $\gamma^* = 5/18$, and the graph $3K_4$ has $T = 12$ and $Q = 3$; so $N_{\leq 10}(3K_4) - N_{\leq 10}(2K_{3,3}) = 21$, the two counts being 4002 and 3981.

5 Small thresholds: proofs of Theorems B and B'

5.1 The switched complete bipartite graph

Proof of Theorem B. Write $K = K_{d,d}$. Each of x_1, x_2, y_1, y_2 loses one edge and gains one, so S_d is d -regular. For $A \subseteq X \cup Y$ put $A_X = A \cap X$, $A_Y = A \cap Y$, $a = |A_X|$ and $b = |A_Y|$, so that $e_K(A) = ab$.

Inclusion. If $e_K(A) \leq 1$, then $a = 0$, or $b = 0$, or $a = b = 1$. If $b = 0$, then $A \subseteq X$, and the only edge of S_d inside X is x_1x_2 ; the case $a = 0$ is symmetric; and a set of two vertices spans at most one edge. So $e_{S_d}(A) \leq 1$. The inclusion is strict: $A_0 = \{x_1, y_1, y_3\}$ has $e_K(A_0) = 2$, but $e_{S_d}(A_0) = 1$, since x_1y_1 was deleted and y_1y_3 is not an edge. (Here $d \geq 3$ is used.)

Exact count. By the inclusion, $N_{\leq 1}(S_d) - N_{\leq 1}(K)$ is the number of sets A with $e_K(A) \geq 2$ and $e_{S_d}(A) \leq 1$. Writing $[\cdot]$ for the indicator of a statement,

$$e_{S_d}(A) = ab - [x_1 \in A_X, y_1 \in A_Y] - [x_2 \in A_X, y_2 \in A_Y] + [x_1, x_2 \in A_X] + [y_1, y_2 \in A_Y].$$

We need $ab \geq 2$ and $e_{S_d}(A) \leq 1$. At most two terms are subtracted, so $ab \leq 3$.

- $(a, b) = (1, 2)$: we need one subtraction and no addition, that is, $A_X = \{x_1\}$ and $y_1 \in A_Y$, or $A_X = \{x_2\}$ and $y_2 \in A_Y$, and in both cases $A_Y \neq \{y_1, y_2\}$. The second vertex of A_Y is then one of $y_3, \dots, y_d$, so there are $2(d-2)$ such sets.
- $(a, b) = (2, 1)$: by symmetry, there are $2(d-2)$ such sets.
- $ab = 3$: then $a = 1$ or $b = 1$, at most one term is subtracted, and $e_{S_d}(A) \geq 2$.

Hence $N_{\leq 1}(S_d) - N_{\leq 1}(K) = 4d - 8$. Finally, $N_{\leq 1}(K) = 2^{d+1} - 1 + d^2$: the independent sets of K are the $2^{d+1} - 1$ subsets of X or of Y , and the sets spanning exactly one edge are the d^2 pairs $\{x, y\}$ with $x \in X$ and $y \in Y$. For $n = 2d$ and $\gamma \in [1/(2d^2), 1/d^2]$ we have $\lfloor \gamma dn \rfloor = 1$, which gives the last assertion. $\square$

Corollary 5. *Let $d \geq 3$ and $m \geq 2$, and put $I = 2^{d+1} - 1$. Then*

$$N_{\leq 1}(S_d \cup (m-1)K_{d,d}) - N_{\leq 1}(mK_{d,d}) = I^{m-2}[(4d-8)I - (m-1)d^2(2^{d-1} - 2)].$$

Hence, for $n = 2dm$ and $\gamma \in [1/(dn), 2/(dn))$, the graph $S_d \cup (m-1)K_{d,d}$ beats $mK_{d,d}$ if and only if $(4d-8)I > (m-1)d^2(2^{d-1} - 2)$. This holds exactly for $d = 3$ with $2 \leq m \leq 4$, for $d \in \{4, 5\}$ with $2 \leq m \leq 3$, and for $6 \leq d \leq 13$ with $m = 2$. For $d \geq 14$ it fails for every $m \geq 2$.

Proof. For disjoint graphs G and F , a set spans at most one edge of $G \cup F$ exactly when it spans at most one edge of G and none of F , or none of G and exactly one of F . Hence

$$N_{\leq 1}(G \cup F) = N_{\leq 1}(G) i_0(F) + i_0(G) (N_{\leq 1}(F) - i_0(F)).$$

Take $F = (m-1)K_{d,d}$, so that $i_0(F) = I^{m-1}$ and $N_{\leq 1}(F) - i_0(F) = (m-1)d^2 I^{m-2}$: one block spans one edge and the others none. Subtracting the identities for $G = S_d$ and $G = K_{d,d}$ and using Theorem B, it remains to show $i_0(K_{d,d}) - i_0(S_d) = 2^{d-1} - 2$, that is, $i_0(S_d) = 3 \cdot 2^{d-1} + 1$. An independent set of S_d inside X is a subset of X not containing both x_1 and x_2 , and there are $3 \cdot 2^{d-2}$ of these; the same holds for Y . An independent set meeting both sides has $a, b \geq 1$; if $a = b = 1$ it must be $\{x_1, y_1\}$ or $\{x_2, y_2\}$, and if $ab \geq 2$ then, by the formula for $e_{S_d}(A)$ above, $e_{S_d}(A) \geq ab - 1 \geq 1$ when $\min(a, b) = 1$, and $e_{S_d}(A) \geq ab - 2 \geq 2$ otherwise. So $i_0(S_d) = 2 \cdot 3 \cdot 2^{d-2} - 1 + 2 = 3 \cdot 2^{d-1} + 1$.

For the list, note that the right-hand side of the criterion increases with m , so for each d the values of m for which it holds form an interval starting at $m = 2$; for $3 \leq d \leq 13$ the list follows by direct evaluation. For $d \geq 14$, the right-hand side minus the left-hand side at $m = 2$ equals $(d^2 - 16d + 32)2^{d-1} - 2d^2 + 4d - 8$, which is positive because $d^2 - 16d + 32 = (d-8)^2 - 32 \geq 4$ and $2^{d+1} > 2d^2 + 8$. $\square$

5.2 A two-block switch

Let $K = K_{d,d}$ and, for $\alpha, \beta \in \{0, 1\}$,

$$W_{\alpha\beta}(z) = \sum_{i,j=0}^{d-1} \binom{d-1}{i} \binom{d-1}{j} z^{ij+\alpha j+\beta i}.$$

Exchanging i and j shows $W_{10} = W_{01}$. Put $D(z) = W_{00}(z)W_{11}(z) - W_{10}(z)^2$.

Lemma 6. *Let $d \geq 2$ and let H_d be as in Theorem B'. Then*

$$P_{2K}(z) - P_{H_d}(z) = 2(z-1)D(z).$$

Consequently, for every graph F , possibly with no vertices, and every integer $t \geq 0$,

$$N_{\leq t}(H_d \cup F) - N_{\leq t}(2K \cup F) = 2[z^t](D(z)P_F(z)).$$

Proof. The graphs $2K$ and H_d have the same vertex set. For A in it and $k = 1, 2$ let $\alpha_k = [u_k \in A]$, $\beta_k = [w_k \in A]$, $i_k = |A \cap (X_k \setminus \{u_k\})|$ and $j_k = |A \cap (Y_k \setminus \{w_k\})|$. The edges of the k -th copy of K other than $u_k w_k$ that lie inside A number $i_k j_k + \alpha_k j_k + \beta_k i_k$, and both graphs contain all of them. In addition, $2K$ contains the edges $u_1 w_1$ and $u_2 w_2$, which contribute $\alpha_1 \beta_1 + \alpha_2 \beta_2$, while H_d contains $u_1 w_2$ and $u_2 w_1$, which contribute $\alpha_1 \beta_2 + \alpha_2 \beta_1$. Summing over A with $(\alpha_1, \beta_1, \alpha_2, \beta_2)$ fixed,

$$P_{2K} = \sum W_{\alpha_1 \beta_1} W_{\alpha_2 \beta_2} z^{\alpha_1 \beta_1 + \alpha_2 \beta_2}, \quad P_{H_d} = \sum W_{\alpha_1 \beta_1} W_{\alpha_2 \beta_2} z^{\alpha_1 \beta_2 + \alpha_2 \beta_1},$$

the sums being over $\{0, 1\}^4$. The two exponents differ by $(\alpha_1 - \alpha_2)(\beta_1 - \beta_2)$, which vanishes unless $\alpha_1 \neq \alpha_2$ and $\beta_1 \neq \beta_2$. The patterns $(\alpha_1, \beta_1, \alpha_2, \beta_2) = (1, 1, 0, 0)$ and $(0, 0, 1, 1)$ each contribute $(z-1)W_{00}W_{11}$ to $P_{2K} - P_{H_d}$, and the patterns $(1, 0, 0, 1)$ and $(0, 1, 1, 0)$ each contribute $(1-z)W_{10}W_{01}$. As $W_{10} = W_{01}$, this proves the identity. For the second assertion, put $R = DP_F$. Since $P_{G \cup F} = P_G P_F$,

$$\begin{aligned} N_{\leq t}(H_d \cup F) - N_{\leq t}(2K \cup F) &= -2 \sum_{s \leq t} [z^s]((z-1)R) \\ &= -2 \sum_{s \leq t} ([z^{s-1}]R - [z^s]R) = 2[z^t]R. \end{aligned} \quad \square$$

Lemma 7. *Let $d \geq 2$ and $q > 0$. Then $D(q) > 0$ if $q > 1$, and $D(q) < 0$ if $q < 1$.*

Proof. For $i, j \in \{0, \dots, d-1\}$ put $\mu(i, j) = \binom{d-1}{i} \binom{d-1}{j} q^{ij}$, and consider

$$\Delta = \sum_{i, i', j, j'} [\mu(i, j) \mu(i', j') - \mu(i, j') \mu(i', j)] (q^i - q^{i'}) (q^j - q^{j'}),$$

where all four indices range over $\{0, \dots, d-1\}$. We have $W_{00}(q) = \sum \mu(i, j)$, $W_{11}(q) = \sum \mu(i, j) q^{i+j}$ and $W_{10}(q) = \sum \mu(i, j) q^j = \sum \mu(i, j) q^i$. Expanding $(q^i - q^{i'}) (q^j - q^{j'})$ into four terms, we find

$$\begin{aligned} \sum_{i, i', j, j'} \mu(i, j) \mu(i', j') (q^i - q^{i'}) (q^j - q^{j'}) &= 2W_{00}W_{11} - 2W_{10}^2, \\ \sum_{i, i', j, j'} \mu(i, j') \mu(i', j) (q^i - q^{i'}) (q^j - q^{j'}) &= 2W_{10}^2 - 2W_{00}W_{11}, \end{aligned}$$

all at the point q . Subtracting, $\Delta = 4D(q)$.

Now let $q \neq 1$. The bracket equals $\binom{d-1}{i} \binom{d-1}{i'} \binom{d-1}{j} \binom{d-1}{j'} q^{ij'+i'j} (q^{(i-i')(j-j')} - 1)$, which has the sign of $(i-i')(j-j') \log q$. Since q^x is strictly monotone in x , the product $(q^i - q^{i'}) (q^j - q^{j'})$ has the sign of $(i-i')(j-j')$. So every term of Δ is zero or has the sign of $\log q$, and the term with $i = j = 0$ and $i' = j' = 1$, which exists because $d-1 \geq 1$, is not zero. $\square$

Proof of Theorem B'. Each of u_1, u_2, w_1, w_2 loses one edge and gains one, so H_d is d -regular. It is bipartite with sides $X_1 \cup X_2$ and $Y_1 \cup Y_2$, since the new edges join X_1 to Y_2 and X_2 to Y_1 . A copy of $K_{d,d}$ minus one edge is connected when $d \geq 2$, and the edge $u_1 w_2$ joins the two copies, so H_d is connected.

By Lemma 6 with F the graph with no vertices ($P_F = 1$), $N_{\leq t}(H_d) - N_{\leq t}(2K_{d,d}) = 2[z^t]D$ for every $t \geq 0$. The exponent in W_{00} is ij , so $[z^0]W_{00} = 2^d - 1$ (the pairs with $i = 0$ or $j = 0$) and $[z^1]W_{00} = (d-1)^2$. The exponent in W_{11} is $(i+1)(j+1) - 1$, so $[z^0]W_{11} = 1$ and $[z^1]W_{11} = 2(d-1)$. The exponent in W_{10} is $j(i+1)$, so $[z^0]W_{10} = 2^{d-1}$ and $[z^1]W_{10} = d-1$. Hence

$$\begin{aligned} [z^0]D &= 2^d - 1 - 4^{d-1} = -(2^{d-1} - 1)^2, \\ [z^1]D &= 2(d-1)(2^d - 1) + (d-1)^2 - 2 \cdot 2^{d-1}(d-1) = (d-1)(2^d + d - 3). \end{aligned}$$

Since $i_0 = N_{\leq 0}$, this proves the two formulas. For $n = 4d$ and $\gamma \in [1/(4d^2), 1/(2d^2))$ we have $\lfloor \gamma dn \rfloor = 1$. $\square$

6 The exponential scale: proof of Proposition C

Let ξ be the number of edges of $K_{d,d}$ spanned by a uniformly random subset of $V(K_{d,d})$. Then ξ takes values in $\{0, \dots, d^2\}$, $\mathbb{P}(\xi = 0) > 0$, $\mathbb{P}(\xi = d^2) = 4^{-d}$, $\mathbb{E}\xi = d^2/4$ because each edge is spanned with probability $1/4$, and $\mathbb{E}q^\xi = 4^{-d}P_d(q)$ for $q > 0$. For $m \geq 1$ let $\Sigma_m = \xi_1 + \dots + \xi_m$, where $\xi_1, \dots, \xi_m$ are independent copies of ξ . A uniformly random subset of $V(mK_{d,d})$ meets the m copies in independent uniformly random subsets. Hence, for $n = 2dm$ and $x = 2\gamma d^2$, so that $\gamma dn = mx$,

$$i_\gamma(mK_{d,d}) = 2^n \mathbb{P}(\Sigma_m \leq mx), \quad 2^n - i_\gamma(mK_{d,d}) = 2^n \mathbb{P}(\Sigma_m > mx). \quad (1)$$

We use the lower bound in Cramér's theorem in the following form.

Lemma 8. *Let ξ be a random variable with values in $\{0, 1, \dots, M\}$ such that $\mathbb{P}(\xi = 0) > 0$ and $\mathbb{P}(\xi = M) > 0$, let Σ_m be the sum of m independent copies of ξ , and let $\Lambda(\lambda) = \log \mathbb{E}e^{\lambda \xi}$.*

(a) If $\mathbb{E}\xi < x < M$, then $\liminf_{m \rightarrow \infty} \frac{1}{m} \log \mathbb{P}(\Sigma_m > mx) \geq \inf_{\lambda \geq 0} (\Lambda(\lambda) - \lambda x)$.

(b) If $0 < x < \mathbb{E}\xi$, then $\liminf_{m \rightarrow \infty} \frac{1}{m} \log \mathbb{P}(\Sigma_m < mx) \geq \inf_{\lambda \leq 0} (\Lambda(\lambda) - \lambda x)$.

Proof. Let $\Lambda^*(y) = \sup_{\lambda \in \mathbb{R}} (\lambda y - \Lambda(\lambda))$. By Cramér's theorem [4, Theorem 2.2.3], for every open set $U \subseteq \mathbb{R}$,

$$\liminf_{m \rightarrow \infty} \frac{1}{m} \log \mathbb{P}(\Sigma_m/m \in U) \geq - \inf_{y \in U} \Lambda^*(y).$$

The function Λ^* is convex, as a supremum of affine functions. It is finite on $[0, M]$: for all λ we have $\Lambda(\lambda) \geq \lambda M + \log \mathbb{P}(\xi = M)$ and $\Lambda(\lambda) \geq \log \mathbb{P}(\xi = 0)$, so for $0 \leq y \leq M$ the quantity $\lambda y - \Lambda(\lambda)$ is at most $-\log \mathbb{P}(\xi = M)$ when $\lambda \geq 0$ and at most $-\log \mathbb{P}(\xi = 0)$ when $\lambda \leq 0$. Hence Λ^* is continuous on $(0, M)$. By Jensen's inequality $\Lambda(\lambda) \geq \lambda \mathbb{E}\xi$, so for $y \geq \mathbb{E}\xi$ the terms with $\lambda < 0$ are at most 0, the value at $\lambda = 0$; thus $\Lambda^*(y) = \sup_{\lambda \geq 0} (\lambda y - \Lambda(\lambda))$, which is nondecreasing in $y \geq \mathbb{E}\xi$. In case (a), take $U = (x, \infty)$; then $\inf_{y > x} \Lambda^*(y) = \lim_{y \downarrow x} \Lambda^*(y) = \Lambda^*(x) = - \inf_{\lambda \geq 0} (\Lambda(\lambda) - \lambda x)$. Case (b) is symmetric, with $U = (-\infty, x)$ and Λ^* nonincreasing on $(-\infty, \mathbb{E}\xi]$. $\square$

Proof of Proposition C. Upper bound. By [3, Proposition 2.1] with $\lambda = 1$, every d -regular graph G on n vertices satisfies $i_\gamma(G) \leq e^{\gamma t n} P_d(e^{-t/d})^{n/(2d)}$ for every $t \geq 0$. With $q = e^{-t/d} \in (0, 1]$ this reads $i_\gamma(G) \leq q^{-\gamma d n} P_d(q)^{n/(2d)}$. Taking logarithms, dividing by n and taking the infimum over q gives $\frac{1}{n} \log i_\gamma(G) \leq \rho(d, \gamma)$. (The proof in [3] combines the bound $P_G(q) \leq P_d(q)^{n/(2d)}$ of [13, Corollary 1.15] with the inequality $[e_G(A) \leq \gamma d n] \leq q^{e_G(A) - \gamma d n}$.)

Lower bound. Let $n = 2dm$, and let $x = 2\gamma d^2$ and Σ_m be as in (1). By the upper bound, it suffices to show that the lower limit of $\frac{1}{n} \log i_\gamma(mK_{d,d})$ as $m \rightarrow \infty$ is at least $\rho(d, \gamma)$.

- $\gamma = 0$: here $i_0(mK_{d,d}) = i_0(K_{d,d})^m$ exactly, so $\frac{1}{n} \log i_0(mK_{d,d}) = \frac{1}{2d} \log i_0(K_{d,d}) = \lim_{q \rightarrow 0} \frac{1}{2d} \log P_d(q) \geq \rho(d, 0)$.
- $0 < \gamma < 1/8$: then $0 < x < d^2/4 = \mathbb{E}\xi$. By (1) and Lemma 8(b), with $q = e^\lambda$,

$$\begin{aligned} \liminf_{m \rightarrow \infty} \frac{1}{m} \log i_\gamma(mK_{d,d}) &\geq 2d \log 2 + \inf_{0 < q \leq 1} (\log(4^{-d} P_d(q)) - x \log q) \\ &= \inf_{0 < q \leq 1} (\log P_d(q) - x \log q). \end{aligned}$$

Dividing by $2d$ and noting $x/(2d) = \gamma d$ gives the claim.

- $\gamma \geq 1/8$: then $x \geq \mathbb{E}\xi$, and $\mathbb{P}(\Sigma_m \leq mx) \geq \mathbb{P}(\Sigma_m \leq m\mathbb{E}\xi) \rightarrow 1/2$ by the central limit theorem, as ξ is not constant. Hence $\frac{1}{n} \log i_\gamma(mK_{d,d}) \rightarrow \log 2$, and $\log 2 \geq \rho(d, \gamma)$ is the value at $q = 1$. $\square$

Remark 9. In the notation of [3], $P_d(e^{-t/d}) = B_d(t)$, so $\rho(d, \gamma) = \inf_{t \geq 0} (\gamma t + \frac{1}{2d} \log B_d(t))$. Carenini's Theorem 1.1 bounds this infimum by choosing $t = 2 \log(e/\gamma)$.

7 Fixed thresholds above one eighth: proof of Theorem D

Let $n = 2dm$, $x = 2\gamma d^2$ and ξ , Σ_m as in Section 6; for $1/8 < \gamma < 1/2$ we have $d^2/4 < x < d^2$. For a graph G on n vertices let $U(G) = 2^n - i_\gamma(G)$, the number of sets spanning more than $\gamma d n$ edges. We must show $U(G_n) < U(mK_{d,d})$ for all large m . Write $H = H_d$.

Proof of Theorem D. Lower bound for $U(mK_{d,d})$. Let $\varphi(q) = \log P_d(q) - x \log q$ for $q > 0$. Since $P_d(q) = 4^d \mathbb{E} q^\xi$, we have $\varphi'(1) = \mathbb{E}\xi - x < 0$; and $\varphi(q) \geq (d^2 - x) \log q \rightarrow \infty$ as $q \rightarrow \infty$, because $P_d(q) \geq q^{d^2}$. Hence φ attains its minimum over $[1, \infty)$ at some $q^* > 1$. By (1) and Lemma 8(a), with $q = e^\lambda$, for every $\varepsilon > 0$ and all large m ,

$$U(mK_{d,d}) = 2^n \mathbb{P}(\Sigma_m > mx) \geq 2^n \exp(m(\varphi(q^*) - d \log 4 - \varepsilon)) = P_d(q^*)^m (q^*)^{-mx} e^{-\varepsilon m}.$$

Upper bound for $U(G_n)$. For $q \geq 1$ and every set A we have $[e(A) > \gamma dn] \leq q^{e(A) - \gamma dn}$. Summing over A with $q = q^*$ gives $U(G_n) \leq (q^*)^{-mx} P_{G_n}(q^*) = (q^*)^{-mx} P_H(q^*)^k P_d(q^*)^r$.

Comparison. As $m - r = 2k$,

$$\frac{U(mK_{d,d})}{U(G_n)} \geq \left(\frac{P_d(q^*)^2}{P_H(q^*)} \right)^k e^{-\varepsilon m}.$$

By Lemma 6 at $z = q^*$ and Lemma 7, $P_d(q^*)^2 - P_H(q^*) = 2(q^* - 1)D(q^*) > 0$, so $\beta = \log(P_d(q^*)^2/P_H(q^*)) > 0$. Take $\varepsilon = \beta/4$. Since $k \geq (m - 1)/2$, the ratio is at least $\exp(\beta(m - 1)/2 - \beta m/4) = \exp(\beta(m - 2)/4) > 1$ for all large m . Hence $U(G_n) < U(mK_{d,d})$, that is, $i_\gamma(G_n) > i_\gamma(mK_{d,d})$, for all large n divisible by $2d$. $\square$

Remark 10. For $\gamma > 1/8$ the law of large numbers gives $\mathbb{P}(\Sigma_m \leq mx) \rightarrow 1$, so $i_\gamma(mK_{d,d}) = (1 - o(1))2^n$, and the relative difference $i_\gamma(G_n)/i_\gamma(mK_{d,d}) - 1 \leq U(mK_{d,d})/i_\gamma(mK_{d,d})$ tends to 0 exponentially fast. For example, for $d = 2$, $\gamma = 1/4$ and $n = 400$ the relative difference is about $1.3 \cdot 10^{-17}$. In particular, Theorem D is consistent with reading R6. Our proof gives no explicit n_0 ; one could be extracted from a quantitative form of Cramér's theorem (for example, exponential tilting combined with the Berry–Esseen theorem, or the theorem of Bahadur and Rao [2]), which we have not done. The base $P_d(q^*)^2/P_H(q^*)$ tends to 1 as $\gamma \downarrow 1/8$, since then $q^* \rightarrow 1$; for $d = 2$ it is about 1.0006 at $\gamma = 1/4$ and 1.0001 at $\gamma = 0.49$. So n_0 can be large, and we give no value.

Reading R6 also holds at $\gamma = 1/8$. Let A be a uniformly random subset of $V(G)$, where G is d -regular on n vertices. Then $e_G(A)$ is the sum of the indicators $[u, v \in A]$ over the $dn/2$ edges uv of G ; it has mean $dn/8$ and variance $dn(2d + 1)/32$, as each indicator has variance $3/16$ and each of the $nd(d - 1)$ ordered pairs of distinct edges with a common end has covariance $1/16$. The indicators of two sets of edges that share no vertex are independent, so the line graph of G , of maximum degree $2d - 2$, is a dependency graph for them, and a normal approximation bound [12, Theorem 3.5 and Proposition 1.2] gives $i_{1/8}(G) = 2^n \mathbb{P}(e_G(A) \leq dn/8) = (1/2 + O(n^{-1/4})) 2^n$, uniformly in G for fixed d .

Remark 11. For $0 < \gamma < 1/8$ the same argument, applied to the lower tail, shows that G_n loses to $mK_{d,d}$ for all large n . Indeed, now $0 < x < \mathbb{E}\xi$, so φ is minimised over $(0, 1]$ at some $q^* < 1$ (note $\varphi'(1) > 0$ and $\varphi(q) \rightarrow \infty$ as $q \rightarrow 0$). By Lemma 7, $D(q^*) < 0$, so again $P_d(q^*)^2 - P_H(q^*) = 2(q^* - 1)D(q^*) > 0$. Lemma 8(b) gives $i_\gamma(mK_{d,d}) \geq P_d(q^*)^m (q^*)^{-mx} e^{-\varepsilon m}$, the inequality $[e(A) \leq \gamma dn] \leq q^{e(A) - \gamma dn}$ for $q \leq 1$ gives $i_\gamma(G_n) \leq (q^*)^{-mx} P_H(q^*)^k P_d(q^*)^r$, and the comparison is as above. This is consistent with Proposition C.

Remark 12. The small- γ failures are nevertheless not confined to tiny graphs. At $\gamma = 1/10$, the graph $G_n = kH_d \cup rK_{d,d}$ of Theorem D beats $mK_{d,d}$ for 92 of the 119 values $n \in \{8, 12, \dots, 480\}$ when $d = 2$, the largest being $n = 400$, and for 56 of the 119 values $n \in \{12, 18, \dots, 720\}$ when $d = 3$, the largest being $n = 360$ (Section 8). By Remark 11 these are finite-size effects.

8 Computations

All counts in this paper were computed in exact integer arithmetic. Our programs compute the distribution $(N_s(G))_s$ in two independent ways: by a recursion over bitmasks, $e_G(A \cup \{v\}) = e_G(A) + |N(v) \cap A|$ for a vertex v larger than every element of A , and by a direct loop over all sets and all edges. The two agree on eleven test graphs. Counts for disjoint unions are obtained from $P_{G \cup F} = P_G P_F$. Two referee agents (see the disclosure at the end of the paper) recomputed the main numbers, each with its own programs. Our programs and their logs are available in version 1.3.0 of the repository [5] (directory `certificates/carenini/`).

Exhaustive search at $n = 2d$. For $d \in \{3, 4, 5\}$ we generated all d -regular graphs on $2d$ vertices up to isomorphism, starting from $K_{d,d}$ and applying 2-switches (replacing two edges ab, cd by ac, bd or by ad, bc when these are not edges), with exact isomorphism tests. Any two graphs with the same degree sequence are connected by a sequence of 2-switches [7, 6], so this finds every class. We found 2, 6 and 60 classes. Every d -regular graph on $2d$ vertices is connected, since each component has at least $d + 1$ vertices, and these are the numbers of connected cubic graphs on 6 vertices, quartic graphs on 8 vertices and quintic graphs on 10 vertices (sequences A002851, A006820 and A006821 of [11]). The first referee agent confirmed the cases $d = 3, 4$ from all 70 and 19355 labelled graphs. Table 3 lists the thresholds at which $K_{d,d}$ is not a maximiser.

d	n	classes	t	$\max_G N_{\leq t}(G)$	$N_{\leq t}(K_{d,d})$
3	6	2	1	28	24
4	8	6	1	61	47
			3	137	127
			5	187	171
5	10	60	1	116	88
			2	203	188
			3	321	288
			5	524	448
			7	679	648
			8	783	748
			11	908	868

Table 3: All thresholds t at which $K_{d,d}$ does not maximise $N_{\leq t}$ among d -regular graphs on $n = 2d$ vertices, for $3 \leq d \leq 5$. The maximum is over all such graphs; the failure holds for $\gamma \in [t/(2d^2), (t+1)/(2d^2))$.

At every other threshold $t \leq d^2$, $K_{d,d}$ is a maximiser. It is the unique maximiser for $t \in \{0, 2\}$ when $d = 3$, for $t \in \{0, 4, 6\}$ when $d = 4$, and for $t \in \{0, 4, 6, 9, 10, 12\}$ when $d = 5$; at the remaining thresholds it ties with other graphs. (By Lemma 1, for $d \geq 3$ all d -regular graphs on n vertices have the same value of $N_{\leq t}$ when $t \geq dn/2 - 2d$.) The failures at $t = 1$ are those of Theorem B, and the failures at $t = d^2 - 3d + 1$, that is, $t = 1, 5, 11$, are those of Theorem A (see also Remark 4); the other failures are not covered by our theorems.

Other checks. The formula of Proposition 2 was checked on 100 regular graphs with $2 \leq d \leq 8$ and $n \leq 20$ (unions of copies of $K_{d,d}$, $(K_d \square K_2) \cup (m-1)K_{d,d}$, cycles, $C_3 \cup C_{n-3}$, unions of copies of K_{d+1} , and random regular graphs), with no mismatch; the first referee agent checked 75 further graphs, including graphs in which 5-sets contribute to Q . Theorem B was checked for $3 \leq d \leq 12$, and Corollary 5 against brute force for $(d, m) \in \{(3, 2), (3, 3), (4, 2), (5, 2)\}$. The identity of Lemma 6 was checked symbolically for $2 \leq d \leq 5$, and the difference in Theorem B' for $2 \leq d \leq 20$; the sign in Lemma 7 was checked on a grid of values of q , as a sanity check. For Theorem D, we compared kH_d with $2kK_{d,d}$ exactly for $d \in \{2, 3, 4\}$ and all $n \leq 600$ divisible by $4d$. At $n = 192$ the competitor wins for $\gamma \in \{1/5, 1/4, 1/3, 2/5, 9/20\}$, while for $\gamma = 3/20$, closer to $1/8$, it still loses, as expected when n_0 is large. The outcome is not monotone in n , so this does not show that $n_0 \leq 192$: for $d = 4$ and $\gamma = 9/20$ the competitor loses at $n = 128$ and again at $n = 208$, and for $d = 3$ and $\gamma = 9/20$ it wins at $n = 96$ and $n = 192$ but loses at $n = 144$. For $\gamma = 3/20$ it wins for every such n from 360 to 600 when $d = 2$, from 336 to 600 when $d = 3$, and from 320 to 592 when $d = 4$, in line with Theorem D. The counts in Remark 12 are exact comparisons for all n in the stated ranges.

9 Formal verification

The Lean file `lean/Research/Carenini.lean` in version 1.3.0 of the repository [5] formalises Question 1.3 literally and proves five negative results: four instances and one infinite family ($d \geq 3$). It uses Lean 4 (version 4.33.1) and Mathlib [10] (commit 0df444a3); all declarations are in the namespace `Carenini`. The basic definitions are the following.

- `edgesIn G A` is the number of edges of G (elements of Mathlib’s `G.edgeFinset`) with both ends in the finite set A ; this is $e_G(A)$. `iCount G t` is the number of sets A with `edgesIn G A` $\leq t$, for $t \in \mathbb{N}$; this is $N_{\leq t}(G)$.
- `iGamma G d gamma` is the number of sets A with `edgesIn G A` $\leq \gamma \cdot d \cdot |V|$, the comparison being made in $\mathbb{R}$ for a real number γ ; this is $i_\gamma(G)$.
- `KddUnion m d` is the graph on `Fin m` $\times$ (`Fin d` $\oplus$ `Fin d`) in which (i, x) and (j, y) are adjacent when $i = j$ and x, y are adjacent in Mathlib’s `completeBipartiteGraph (Fin d) (Fin d)`; this is $mK_{d,d}$.

The question and the two main theorems read, in ASCII notation:

```
def CareniniQuestion (n d : Nat) (gamma : Real) : Prop :=
  forall (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V)
    [DecidableRel G.Adj], Fintype.card V = n -> G.IsRegularOfDegree d ->
    iGamma G d gamma <= iGamma (KddUnion (n / (2 * d)) d) d gamma
```

```
theorem not_careniniQuestion_six : Not (CareniniQuestion 6 3 (1 / 18))
```

```
theorem not_careniniQuestion_two_d {d : Nat} (hd : 3 <= d) :
  Not (CareniniQuestion (2 * d) d (1 / (2 * (d : Real) ^ 2)))
```

The predicate quantifies over every finite vertex type of cardinality n and every d -regular simple graph on it (Mathlib’s `IsRegularOfDegree`); the threshold γdn is a real number, as in the question; and the reference graph consists of $n/(2d)$ disjoint copies of Mathlib’s $K_{d,d}$. The standing hypothesis $2d \mid n$ is not part of the predicate. Every instance refuted below satisfies it, and at such an instance the predicate is exactly the claim of Question 1.3. The lemma `iGamma_eq_iCount_floor` shows that $i_\gamma = N_{\leq \lfloor \gamma dn \rfloor}$ for $\gamma \geq 0$, and `iGamma_iso` shows that i_γ is invariant under isomorphism. Table 4 lists the five negative results: four instances and one infinite family ($d \geq 3$).

Lean theorem	(n, d, γ)	competitor and counts
<code>not_careniniQuestion_six</code>	$(6, 3, 1/18)$	prism, $28 > 24$ (Theorems A, B)
<code>not_careniniQuestion_two_d</code>	$(2d, d, 1/(2d^2)), d \geq 3$	S_d (Theorem B)
<code>not_careniniQuestion_eight_two</code>	$(8, 2, 1/16)$	$C_8 = H_2$, $111 > 105$ (Theorem B')
<code>not_careniniQuestion_twelve_bip</code>	$(12, 3, 1/36)$	H_3 , $527 > 495$ (Theorem B')
<code>not_careniniQuestion_twelve_top</code>	$(12, 3, 5/18)$	$3K_4$, $4002 > 3981$ (Theorem A)

Table 4: The negative results proved in Lean. In each case the reference graph is $\frac{n}{2d}K_{d,d}$.

Supporting declarations make the statements concrete. For $n = 6$, `prism_regular` and `K33_regular` are proved by `decide`, the counts 28 and 24 are evaluated by `decide`, and `prismIso`, `K33Iso` and `K33IsoKddUnion` show that the two graphs are Mathlib’s $K_3 \square K_2$ and $K_{3,3}$, the latter also as `KddUnion 1 3`; `prism_beats_K33` is the explicit form of the instance. For general d , `switchedKdd_regular` shows that S_d is d -regular, `edgesIn_switched_le_one` is the inclusion in Theorem B, `switched_extra_set` is the witness $\{x_1, y_1, y_3\}$ (indexed from 0 in

Lean), `iCount_one_lt` gives $N_{\leq 1}(K_{d,d}) < N_{\leq 1}(S_d)$, and `kddUnionOneIso` identifies $K_{d,d}$ with `KddUnion 1 d`. Finally, `H3_regular`, `H3_bipartite` and `H3_connected` prove the properties of H_3 stated in Theorem B'.

The following are not formalised: Theorem A (only two instances of its conclusion are formal, the prism at $(6, 3, 1/18)$ and $3K_4$ at $(12, 3, 5/18)$; Proposition 2, Lemma 3 and the minimiser statement are not); the exact value $4d - 8$ in Theorem B (only the strict inequality is) and Corollary 5; Theorem B' (only $N_{\leq 1}(H_d) > N_{\leq 1}(2K_{d,d})$ for $d = 2, 3$, via the counts $111 > 105$ and $527 > 495$, the regularity of $H_2 = C_8$ and of H_3 , and the bipartiteness and connectivity of H_3 are formal), and Lemmas 6 and 7; Proposition C and Theorem D, which use Cramér's theorem and, for the upper bound in Proposition C, the theorem of Sah, Sawhney, Stoner and Zhao; and the computations of Section 8.

The file contains no `sorry` and does not use `native_decide`; the finite counts are evaluated by the kernel, using `decide` and `decide +kernel`. The file was rebuilt with `lake build` in a clean directory, with exit code 0, and the compiled module was checked again with `leanchecker`. The recorded axiom audits (`#print axioms`) of the fourteen main declarations report only the standard axioms `propext`, `Classical.choice` and `Quot.sound`, and no axioms at all for `H3_bipartite`. An audit file, `lean/Research/CareniniAudit.lean`, prints the basic definitions and the statements of Table 4 together with their axioms. The Lean file, the audit file and the logs of the clean rebuild and of the `leanchecker` run (files `logs/clean-replay-v1.3.0-*`) are in version 1.3.0 of the repository [5].

10 Open questions

1. (Reading R5.) Let $d \geq 2$ and $0 < \gamma \leq 1/8$ be fixed. Is $i_\gamma(G) \leq i_\gamma(mK_{d,d})$ for every d -regular graph G on n vertices, for all sufficiently large n divisible by $2d$? At $\gamma = 1/10$ the inequality fails for many n up to 400 (Remark 12), but the competitors used there lose for large n (Remark 11). For $\gamma < 1/8$ the infimum defining $\rho(d, \gamma)$ is attained at some $q < 1$, in the antiferromagnetic range in which the theorem of Sah, Sawhney, Stoner and Zhao favours $K_{d,d}$. This suggests a positive answer, but we make no claim.
2. All our counterexamples with $\gamma \leq 1/8$ have $\gamma d < 1$. Is there a counterexample, for some n , with $\gamma \leq 1/8$ and $\gamma d \geq 2$, the range of small γ in Carenini's Proposition 1.2?
3. (Reading R6.) For fixed d and $0 < \gamma < 1/8$, is the maximum of i_γ over d -regular graphs on n vertices at most $(1 + o(1)) i_\gamma(mK_{d,d})$ as $n \rightarrow \infty$? (For $\gamma = 1/8$ it is; see Remark 10.)
4. Which d -regular graphs maximise i_γ when $1/8 < \gamma < 1/2$ and n is large? By Proposition 2, at $\gamma = \gamma^*$ the maximisers are the graphs that maximise $2T(G) - Q(G)$.

Tool and computational resource disclosure

This work was carried out on 26 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. The author set the research programme, namely resolving open problems with AI agents and formal verification, and directed the agents; the agents selected this question. Claude Code agents found the counterexamples and the proofs, wrote the verification programs and the Lean formalisation, carried out the literature searches described in the introduction, and drafted this text. An independent Claude Code agent, without access to the reasoning of the agents that found the proofs, refereed the proofs and the correspondence between the Lean statements and the question, and recomputed the main numbers with its own programs. A second independent Claude Code agent then reviewed the whole paper: it checked the proofs and the statements

against the Lean file, recomputed the numbers with its own programs, and checked the references. Before release, OpenAI Codex (desktop application) carried out a brief independent check of the question, the main statements and their Lean formalisation, and recounted the numbers 28 and 24; its one finding, that Theorem A must require $n > 0$, has been applied. The programs, logs and Lean code are available in version 1.3.0 of the repository [5] (directories `certificates/carenini/`, `lean/Research/` and `logs/`). The author is not a professional mathematician. The author has read the paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. The correctness of the results rests on the written proofs and, for the results listed in Table 4, also on the Lean formalisation; the computations of Section 8 are additional cross-checks. AI systems are not authors of this paper; the author takes full responsibility for its content.

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